lecture03b_SimpleModelsOfGeneRegulation

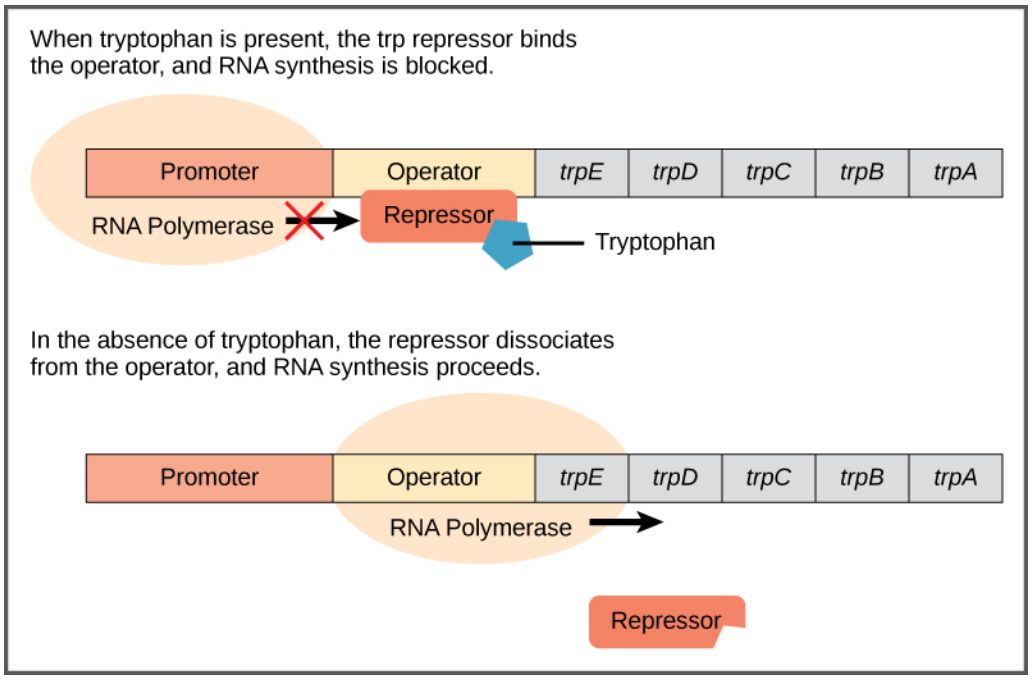
September 30, 2026

1 Gene regulation

The expression of genes is tightly regulated in space and time. * Transcription factors: proteins that bind to DNA and enhance or suppress expression * Sigma factors (transcription initiation) * Chromatin state: the genome can be packaged away and silenced * DNA modifications: methylation, histone marks * Protein modifications, dimerizations, etc

The molecular biology of these process is very complicated and we will focus here on general aspects of the problem.

1.0.1 Typical cartoon of gene regulation



source: bio.libretexts.org/Introductory_and_General_Biology

1.1 Transcription factor binding

- a single TF binds to a binding site according to simple mass-action kinetic:

$$P(\text{TF bound}) = \frac{[X]}{k + [X]}$$

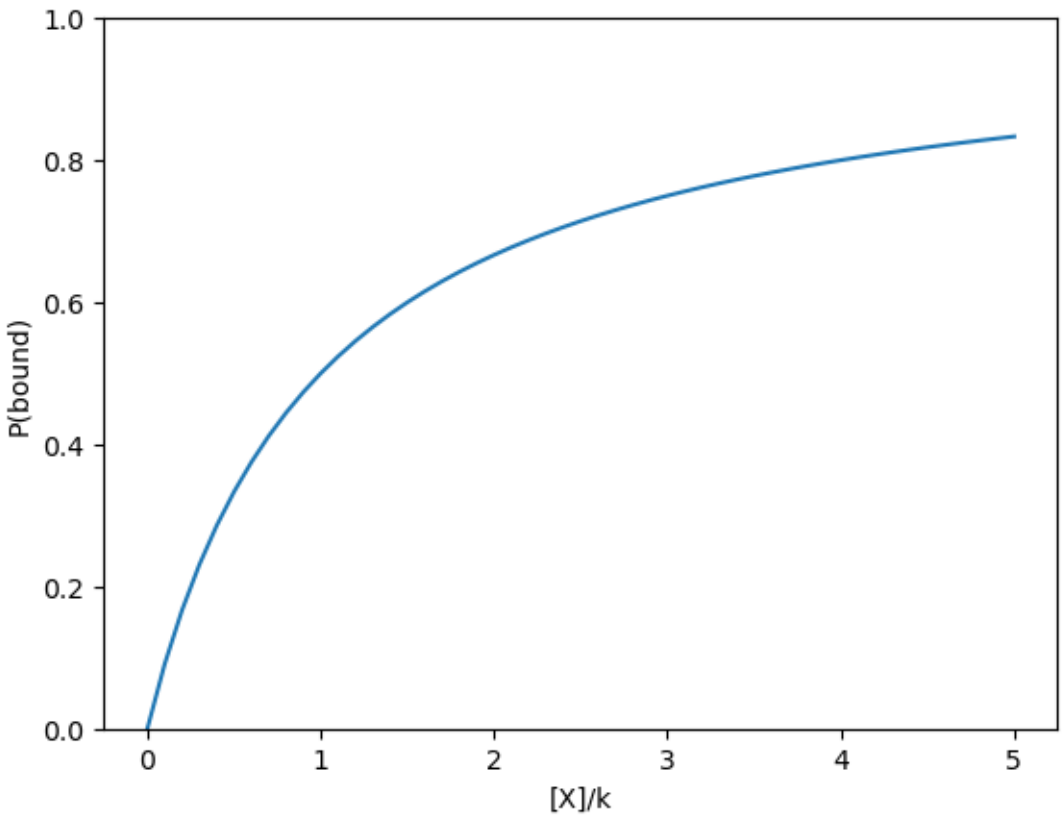
where $[X]$ is the concentration and K the affinity.

- binding is mediated by non-covalent interactions, so k will depend on salt concentrations
- non-cooperative binding: $P(\text{TF bound})$ increases linearly with $[X]$ and saturates at 1 for $[X] \gg k$

[18]:
```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0,5,51)
plt.plot(x, x/(1+x))
plt.ylabel('P(bound)')
plt.xlabel('[X]/k')
plt.ylim([0,1])
```

[18]: (0.0, 1.0)



1.2 Cooperative binding

- precise regulation requires sharper threshold than the simple first order binding above
- often binding is cooperative and involves multiple TFs (e.g. homo- or hetero-dimers)
- similar to co-operative binding to of oxygen to hemoglobin – this will be discussed in more

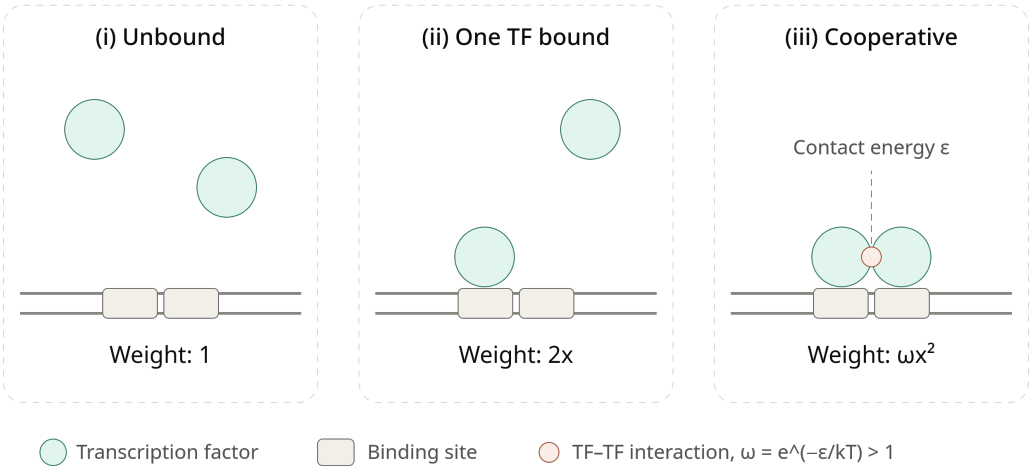
detail in the second part of the lecture by Sebastian Hiller.

Starting from the Michaelis-Menten binding probabilities above, we argued that binding of a single site can be expressed as $P(\text{TF bound}) = \frac{[X]}{k+[X]}$. If now n TFs need to bind simultaneously to activate transcription, it is reasonable to assume that the probability can be expressed as

$$P(\text{all n TF bound}) = \frac{[X]^n}{(k+[X])^n}$$

At very high concentrations, this probability approaches 1, but full occupancy of multiple binding sites by independently binding TFs is quite hard to achieve. To generate more "switch-like" behavior, we require cooperative rather than independent binding.

For simplicity, consider binding of two TFs to adjacent binding sites as illustrated below



In the independent binding case discussed above, we would have as probability for the double-bound state.

$$P(\text{both TF bound}) = \frac{[X]^2}{(k+[X])^2} = \frac{[X]^2}{k^2+2K[X]+[X]^2}$$

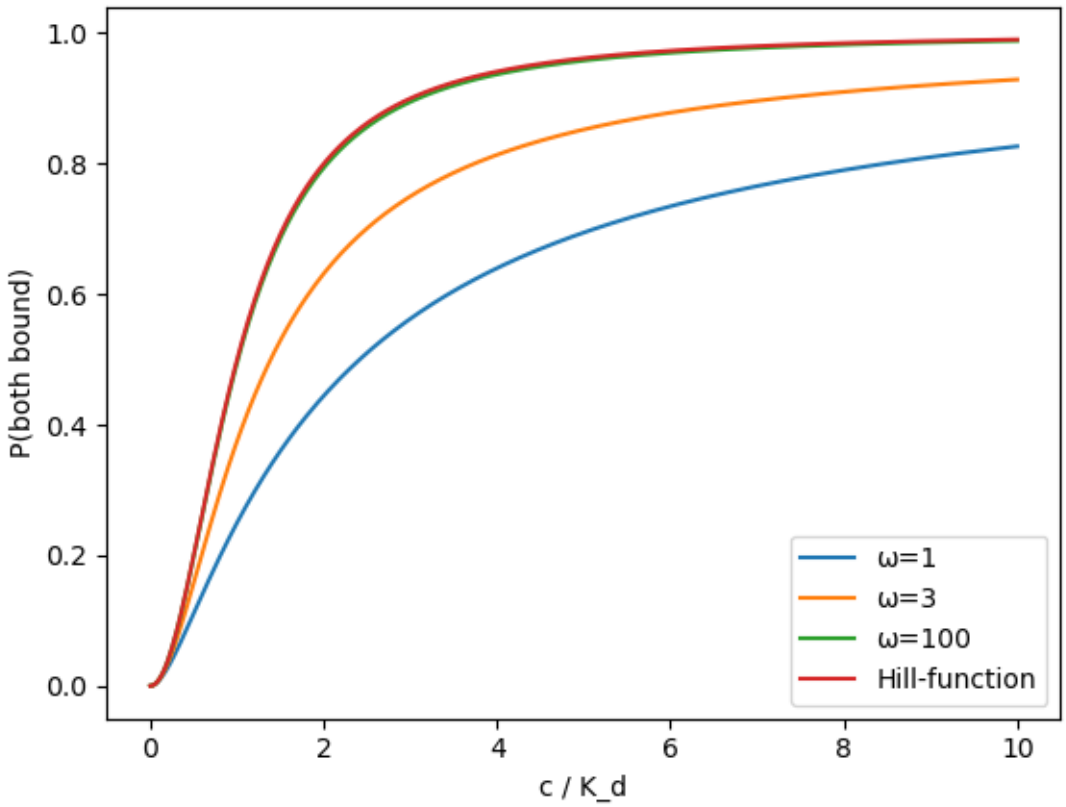
The term in the numerator is proportional to the event of two TFs being bound, the numerator is the sum over all possible states. Their ratio expresses a probability between 0 and 1.

Now of the two TFs don't bind independently but also interact with each other when bound, the double bound state can be enriched relative to the singly bound case. This is parameterized by ω in the illustration above. For substantial interaction energies, we can ignore the singly bound state and approximate cooperative binding as

$$P(\text{both TF bound}) = \frac{[X]^2}{k_e^2+[X]^2}$$

Where we have introduced a modified $k_e = k/\sqrt{w}$ to account for the fact that the overall occupancy is increasing with stronger interactions.

[19]:
```
x = np.linspace(0, 10, 400)
for w in [1, 3, 100]:
    Z = 1*w + 2*x + w*x**2
    plt.plot(x, w*x**2/Z, label=f" ={w}")
plt.plot(x, x**2/(1+x**2), label="Hill-function")
plt.xlabel("c / K_d"); plt.ylabel("P(both bound)"); plt.legend(); plt.show()
```



1.3 Thermodynamic explanation borrowing from Hiller's script

Consider transcription factors A and B

- TF-DNA interaction A: ϵ_A
- TF-DNA interaction B: ϵ_B
- TF-TF interaction: J_{AB}

The grand partition function is then (comp. Hiller's script)

$$\mathcal{Z} = 1 + ([A]/k_A)e^{-\epsilon_A/kT} + ([B]/k_B)e^{-\epsilon_B/kT} + ([A][B]/k_A k_B)e^{-(\epsilon_A+\epsilon_B+J_{AB})/kT} \tag{1}$$

Multiplying by $k_A k_B$, we find for the probability that both A and B are bound:

$$P_{AB} = \frac{[A][B]e^{-(\epsilon_A+\epsilon_B+J_{AB})/kT}}{k_A k_B + \cdots + [A][B]e^{-(\epsilon_A+\epsilon_B+J_{AB})/kT}} \tag{2}$$

If cooperation is strong, this simplifies to:

$$P_{AB} \approx \frac{[A][B]e^{-(\epsilon_A+\epsilon_B+J_{AB})/kT}}{k_A k_B + [A][B]e^{-(\epsilon_A+\epsilon_B+J_{AB})/kT}} = \frac{[A][B]}{K + [A][B]} \tag{3}$$

This form of cooperative binding generalizes to m copies of A and m copies of B as follows.

$$P_{AB} \approx \frac{[A]^m[B]^n}{K + [A]^m[B]^n} \tag{4}$$

1.3.1 Alternative formulation

In the above derivation, the binding of a single TF was characterized by two parameters k_A and ϵ_A. This is a redundant parameterization since we could redefine k_A and k_B as follows

$$k_A \leftarrow k_A e^{-\epsilon_A/kT}$$

and

$$k_B \leftarrow k_B e^{-\epsilon_B/kT}$$

This would allow us to write the partition function as

$$\mathcal{Z} = 1 + [A]/k_A + [B]/k_B + \frac{[A][B]e^{-J_{AB}/kT}}{k_A k_B}$$

(with the redefinition of the parameters, this is exactly the same expression).

The probabilitiy that both A and B are bound is then given by

$$\begin{aligned} P(A\&B) &= \frac{[A][B]}{k_A k_B e^{J_{AB}/kT} + [A]k_B e^{J_{AB}/kT} + [B]k_A e^{J_{AB}/kT} + [A][B]} \\ &\approx \frac{[A][B]}{k_A k_B e^{J_{AB}/kT} + [A][B]} \end{aligned}$$

1.4 Hill-functions and cooperativity

Cooperative binding by a single species is often parameterized by so the so called Hill coefficient n. The probability of activation if the activator is at concentration x is then

$$h(x) = \frac{x^n}{K + x^n}$$

The higher the Hill-coefficient n, the more step-like the function is. $K^{1/n} = k_e$ marks the value at which $h(x) = 0.5$

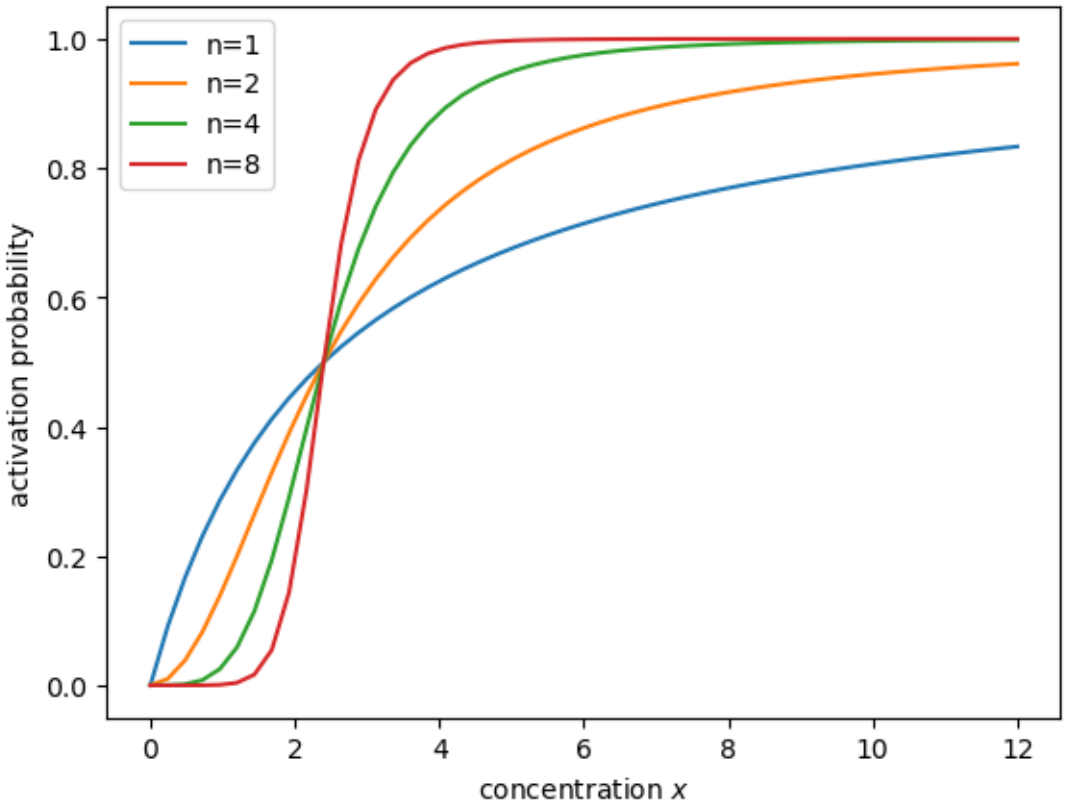
[20]:
```
import numpy as np
import matplotlib.pyplot as plt

k = 2.4
x = np.linspace(0,5*k,51)

for n in [1,2,4,8]:
    K = k**n
    plt.plot(x, x**n/(K+x**n), label=f"n={n}")

plt.legend()
plt.ylabel("activation probability")
plt.xlabel("concentration $x$")
```

[20]:
```
Text(0.5, 0, 'concentration $x$')
```



1.4.1 Repression

The above discussion considered binding of activators. If instead the role of the transcription factor is repression, the expression of the gene is given by 1 minus the binding probability.

$$g(x) = 1 - \frac{x^n}{K + x^n} = \frac{K}{K + x^n}$$

1.5 Why do we need highly cooperative regulation?

Large Hill coefficients are required for sharp responses to minor changes in input variables. This is well illustrated by the response to a network to an external signal.

Assume a quantity x is changing according to an signal in a time dependent manner and y is responding to the input x. Low cooperativity will result in a gradual response, while high cooperativity results in a step wise response.

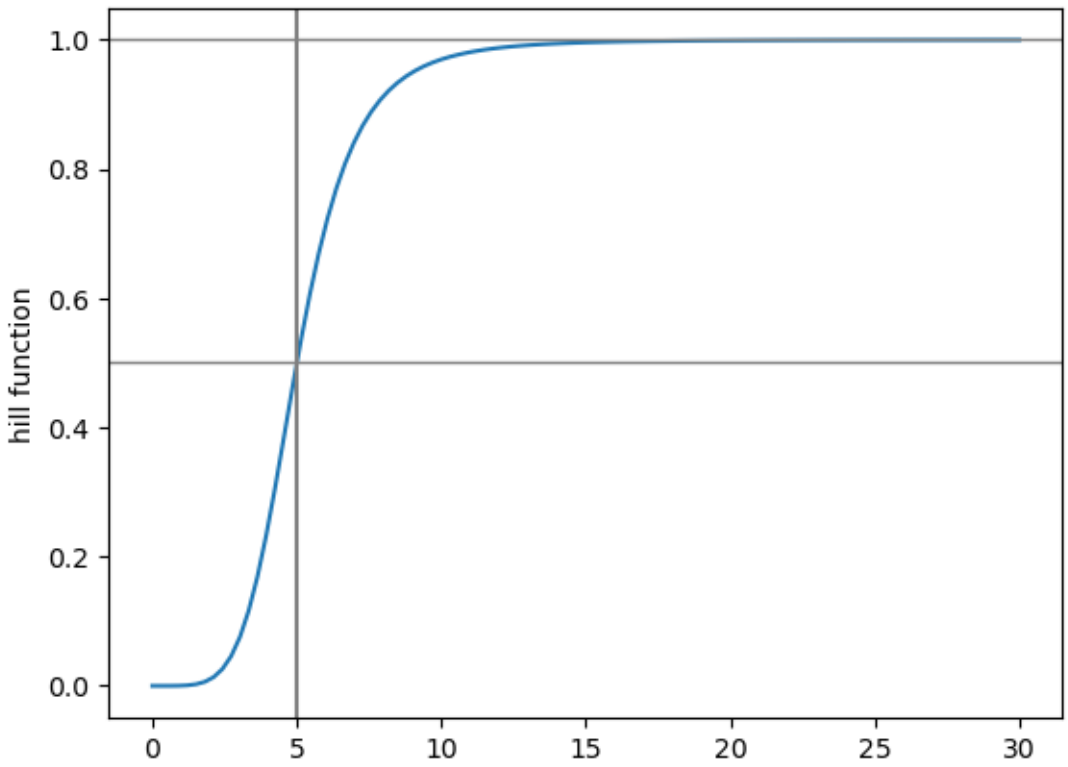
$$\frac{dy}{dt} = \alpha \frac{x^n}{k^n + x^n} - \beta y$$

[26]:
```
from scipy.integrate import odeint
import matplotlib.pyplot as plt
import numpy as np

# define a hill function with cooperativity n
def hillfunc(x, k, n):
    return x**n/(k**n+x**n)


# plot the hill function and indicate its 1/2 value and max value
xvals = np.linspace(0,30,100)
plt.plot(xvals, hillfunc(xvals, 5,5))
plt.axvline(5, c='grey')
plt.axhline(.5, c='grey', lw=1)
plt.axhline(1, c='grey', lw=1)
plt.ylabel('hill function')
```

[26]:
```
Text(0, 0.5, 'hill function')
```



[27]:
```
# define RHS of ODE governing production, note the explicit time dependence!
def dydt(y, t, k, n, alpha, beta):
    #case 1: x high first, low later
    if t<10:
        x=8
    elif t<15:
        x=2
    else:
        x=20
    #case 2: sinusoidal fluctuation
    # x = 5*(1+np.sin(t*2*np.pi/40))

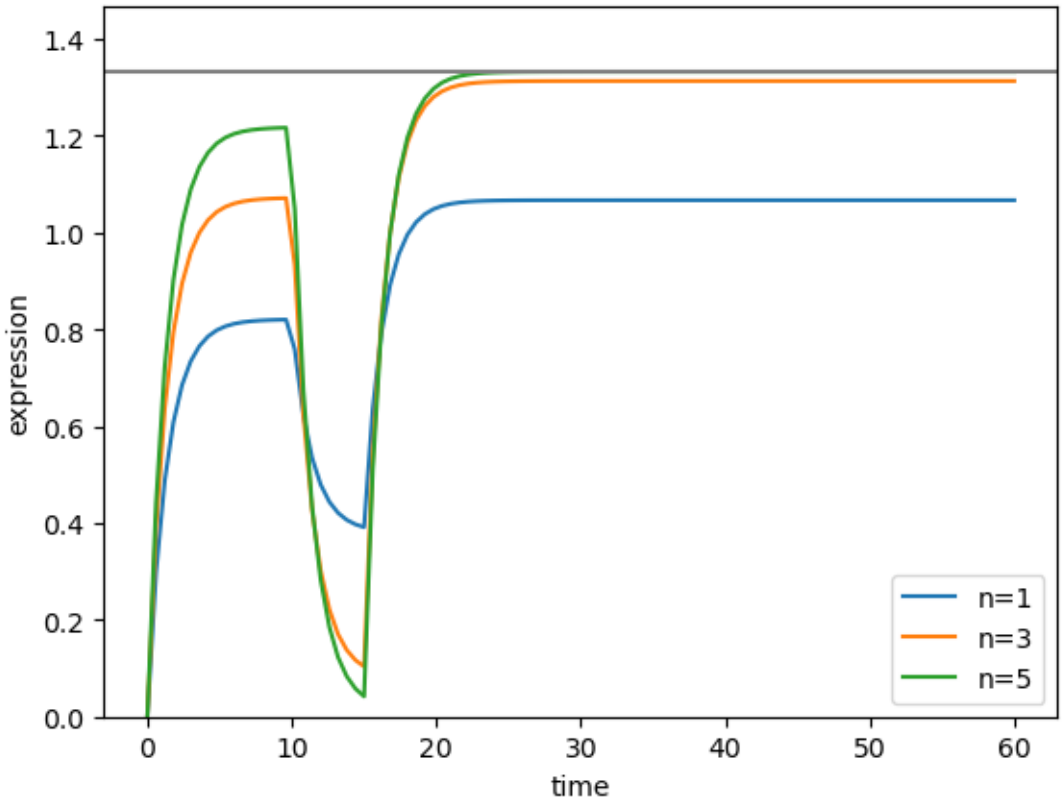
    return alpha*hillfunc(x, k, n) - beta*y


# Solve the solution to the ODE and graph the result
y0 = 0
alpha=1
beta=0.75

for n in [1,3,5]:
    k=5
```

```
    # array of time points to solve the ODE on
    T = np.linspace(0,60,101)

    sol = odeint(dydt, y0, T, (k, n, alpha, beta))
    plt.plot(T, sol[:,0], label=f"n={n}")
plt.axhline(alpha/beta, c='grey') # equilibrium at full expression
plt.xlabel('time')
plt.legend()
plt.ylabel('expression')
plt.ylim(0,alpha/beta*1.1);
```



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